Toy Model 5G-M - Full SPARC Run Results

Status: full 175-galaxy SPARC execution completed using the uploaded Rotmod_LTG.zip. This is not a physical validation claim.

1. What was run

The uploaded Rotmod_LTG.zip archive was staged locally and the frozen 5G-M harness was run across 175 galaxies and 3391 radial points.

The target remained the 5G-L metric distortion object:

```
chi(R) = ln[g_obs(R) / g_bar(R)]
```

No dark halo terms, no per-galaxy fitting, and no post-failure feature changes were introduced.

2. Headline results

Mode	R2 chi	RMSE chi	Global RMS 5G-M	Global RMS Newtonian	Improved galaxies
frozen_four	0.4613	0.5216	36.46 km/s	58.57 km/s	139 / 175
train_groupkfold	0.5669	0.4676	24.69 km/s	58.57 km/s	155 / 175

3. Interpretation

The full-SPARC run supports the decision to move from direct acceleration patches toward metric / connection reconstruction. Both modes improve the global RMS relative to the Newtonian baryonic baseline, and the group-holdout mode performs substantially better than the original frozen four-galaxy extrapolation.

However, this is still not validation. The frozen_four extrapolation is only moderately successful, systematic outliers remain, and the train_groupkfold mode still trains on the larger SPARC population. The correct claim is diagnostic progress, not physical proof.

4. Negative residual structure

The run found 218 points where g_bar exceeds g_obs. This supports the earlier 5G-H lesson: the residual cannot be modeled as a simple positive dark-sector acceleration field. A metric / connection object is the correct mathematical target to continue testing.

5. Best and worst examples

Best group-holdout galaxies by RMS

Galaxy	5G-M RMS	Newton RMS	Neg pts
UGCA281	0.90	10.40	0.00
D512-2	1.11	17.69	0.00
F583-4	2.14	28.26	0.00
D564-8	2.24	11.65	0.00
DDO064	2.69	17.87	0.00
UGCA444	2.74	14.47	0.00
UGC07690	3.41	22.75	0.00
UGC07866	3.57	11.08	0.00

Worst group-holdout galaxies by RMS

Galaxy	5G-M RMS	Newton RMS	Neg pts
NGC5985	62.78	135.98	0.00
NGC5371	58.04	45.31	0.00
NGC4217	55.85	47.86	8.00
UGC02487	54.89	167.75	0.00
NGC6195	52.62	32.47	3.00
UGC11914	49.94	53.53	2.00
NGC4088	49.58	25.40	3.00
NGC4389	49.47	23.57	5.00

6. Publication wording guardrail

Allowed: 5G-M completed a full-SPARC diagnostic run and found that relational metric reconstruction improves strongly over the Newtonian baryonic baseline while leaving important failures.

Not allowed: V4.2 solved dark matter, replaced Lambda-CDM, or achieved physical validation.

7. Next responsible step

Freeze the 5G-L/5G-M target and feature family in a public protocol, then run a stricter pre-registered split over the full SPARC database with independent review of outliers and sensitivity to mass-to-light assumptions.